

$\therefore$ The condition that each row of a matrix has a leading element in a column to the right of the columns of the leading elements of all rows above it is True. $\square$

Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0

$\therefore$ The condition that each pivot column contains zero in all positions below the pivot position is True. $\square$

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in echelon form.

There are no rows in the matrix of the form $[0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. By Theorem 2, the system of equations is consistent, i.e. there exists a solution set. $\therefore b$ is in W

$\begin{bmatrix} 6 \\ 5 \\ 1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 8 \\ -2 \end{bmatrix} + 1 \cdot \begin{bmatrix} 6 \\ 5 \\ 1 \end{bmatrix}$. 0 and 1 are real numbers

$\therefore \begin{bmatrix} 6 \\ 5 \\ 1 \end{bmatrix}$ is a linear combination of the columns of A .

$\therefore \begin{bmatrix} 6 \\ 5 \\ 1 \end{bmatrix}$ is in W ; the third column is in W .

27) A mining company has two mines. One day's operation at mine #1 produces one that contains 20 metric tons of copper and 550 kilograms of silver, while one day's operation at mine #2 produces one that contains 30 metric tons of copper and 500 kilograms of silver. Let $v_1 = \begin{bmatrix} 20 \\ 550 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 30 \\ 500 \end{bmatrix}$. Then v_1 and v_2 represent the "output per day" of mine #1 and mine #2, respectively.

a) What physical interpretation can be given to the vector $5v_1$?

Solution: $5v_1$ represents the total output of 5 days for mine #1. (By scalar multiplication)

$5v_1 = 5 \begin{bmatrix} 20 \\ 550 \end{bmatrix} = \begin{bmatrix} 5 \cdot 20 \\ 5 \cdot 550 \end{bmatrix} = \begin{bmatrix} 100 \\ 2750 \end{bmatrix}$

b) Suppose the company operates mine #1 for x_1 days and mine #2 for x_2 days. Write a vector equation whose solution gives the number of days each mine should operate in order